

SOC/SIEM Incident Report

Microsoft Defender EICAR Detection

Wazuh SIEM | Windows Defender Operational Logs | Endpoint Malware-Test Detection

Prepared for	Cybersecurity Home Lab Portfolio
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Report type	Controlled lab incident report / portfolio case study
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Classification	Lab simulation - safe EICAR antivirus test file; no real malware used

Report Purpose

- Demonstrate endpoint malware-test detection using Microsoft Defender and Wazuh.
- Show the complete workflow from controlled test-file creation through alert investigation and cleanup validation.
- Document evidence in a professional incident-report format suitable for portfolio review.

Executive Summary

A safe EICAR antivirus test file was created on WIN11-ENDPOINT to validate endpoint detection and SIEM log ingestion. Microsoft Defender Antivirus detected the test file in real time, classified it as Virus:DOS/EICAR_Test_File, and quarantined it. The local Windows Defender Operational log recorded Event ID 1117, and Wazuh ingested the event from WIN11-ENDPOINT. The event details in Wazuh confirmed the Defender log channel, event ID, threat name, severity, and endpoint source.

The test was performed in a controlled homelab environment and did not use real malware. Cleanup was verified by confirming the test file no longer existed at C:\Tools\leicar.com.txt and that Microsoft Defender reported a successful action.

Key Findings

- Microsoft Defender detected the EICAR test file through Real-Time Protection.
- The local Defender event showed Event ID 1117 and threat name Virus:DOS/EICAR_Test_File.
- The file path involved was C:\Tools\leicar.com.txt.
- The action taken was quarantine, with no additional actions required.
- Wazuh ingested the Defender Operational event from WIN11-ENDPOINT and generated an event with rule ID 62124.
- Cleanup was validated with Test-Path returning False and Defender action success showing True.

Environment

Component	Details
Hypervisor	Microsoft Hyper-V
SIEM Platform	Wazuh
SIEM Server	Wazuh-SIEM on Ubuntu Server
Wazuh Manager IP	192.168.56.10
Endpoint	WIN11-ENDPOINT
Endpoint IP	192.168.56.20
Endpoint Security Tool	Microsoft Defender Antivirus
Log Sources	Microsoft-Windows-Windows Defender/Operational, Windows EventChannel, Wazuh Agent
Test Artifact	Safe EICAR antivirus test file created in C:\Tools

Scenario Objective

The objective of this scenario was to validate that a Windows endpoint security event generated by Microsoft Defender could be detected locally, forwarded to Wazuh, and investigated from the SIEM. The EICAR test file was used because it is designed for safe antivirus testing and does not represent real malware execution.

Detection Workflow Timeline

Step	Action	Evidence Produced
1	Added Microsoft Defender Operational log collection to the Wazuh agent configuration and restarted the agent.	Wazuh agent prepared to ingest Defender events.
2	Created the safe EICAR test file in C:\Tools using PowerShell.	Windows Security notification appeared showing threats found.
3	Reviewed Microsoft Defender Operational logs locally.	Event ID 1117 showed Defender took action on Virus:DOS/EICAR_Test_File.
4	Searched Wazuh Threat Hunting events for Defender event ID 1117.	Wazuh displayed a Defender event from WIN11-ENDPOINT with rule ID 62124.
5	Verified cleanup and Defender action status.	Test-Path returned False and Get-MpThreatDetection showed ActionSuccess=True.

Evidence Summary

Evidence	Source	Key Fields	Result
EICAR file creation and Defender notification	PowerShell + Windows Security notification	C:\Tools\leicar.com.txt; threats found notification	Defender detected the test file immediately.
Local Defender event	Microsoft-Windows-Windows Defender/Operational	Event ID 1117; Virus:DOS/EICAR_Test_File; Severity=Severe; Action=Quarantine	Defender recorded a successful protective action.
Wazuh event overview	Wazuh Threat Hunting	Rule ID 62124; WIN11-ENDPOINT; Defender action rule description	Wazuh received and displayed the Defender event.
Wazuh event details	Wazuh document details	Channel=Microsoft-Windows-Windows Defender/Operational; Event ID=1117; Threat name=Virus:DOS/EICAR_Test_File	SIEM evidence tied the Defender event to the monitored endpoint.
Cleanup verification	PowerShell + Defender history	Test-Path=False; ActionSuccess=True	Test file was removed/quarantined successfully.

Technical Details

Wazuh Agent Log Collection Added

The Wazuh agent on WIN11-ENDPOINT was configured to collect Microsoft Defender Operational events. This enabled Defender telemetry to be forwarded into Wazuh for alert review and investigation.

```
<localfile>
  <location>Microsoft-Windows-Windows Defender/Operational</location>
  <log_format>eventchannel</log_format>
</localfile>
```

Controlled Test Activity

A safe EICAR antivirus test file was written to C:\Tools\eicar.com.txt using PowerShell. Microsoft Defender detected the file immediately through real-time protection. The exact test string is intentionally not repeated as report text; the evidence screenshot shows the controlled command execution and Defender notification.

Investigation Notes

- Confirmed Microsoft Defender generated a user-facing security notification after the test file was created.
- Queried the Defender Operational log and confirmed Event ID 1117 for Virus:DOS/EICAR_Test_File.
- Confirmed the local Defender event identified the source as Real-Time Protection and action as Quarantine.
- Searched Wazuh Threat Hunting events for Event ID 1117 and confirmed the Defender event appeared under WIN11-ENDPOINT.
- Opened the Wazuh document details to validate the event channel, endpoint, event ID, threat name, and Wazuh rule description.
- Verified cleanup by confirming the test file no longer existed and Defender action success was True.

Potential Impact

In a real environment, a Defender detection involving malware or potentially unwanted software could indicate an attempted download, execution, or staging of malicious content. The severity and response would depend on the detection type, affected host, user context, file path, and whether the security tool successfully remediated the threat.

Containment and Remediation

Microsoft Defender automatically quarantined the EICAR test file. The following cleanup checks were performed:

```
Test-Path C:\Tools\eicar.com.txt
Get-MpThreatDetection | Select-Object -First 3 ThreatName,ActionSuccess,Resources,InitialDetectionTime
```

The file-existence check returned False, and Microsoft Defender threat history showed ActionSuccess=True for the EICAR detection. No additional manual cleanup was required.

Detection Mapping and Notes

This scenario primarily validates endpoint protection telemetry, Defender event collection, and SIEM investigation workflow. The Wazuh event did not assert a specific adversary technique in the screenshot evidence. For reporting accuracy, this document does not claim a specific MITRE ATT&CK technique. The relevant detection concept is endpoint malware detection and protective action validation.

Lessons Learned

- Endpoint security tools may generate valuable operational events that should be forwarded to the SIEM for centralized review.
- A local endpoint detection should be validated both on the endpoint and in the SIEM to prove end-to-end visibility.
- Defender Event ID 1117 is useful evidence that Defender took action against a detected threat.
- Cleanup verification should include both file-state checks and security-tool action status, not only the alert itself.
- Using safe test artifacts allows detection validation without introducing real malware into the lab.

Skills Demonstrated

- Wazuh log source configuration
- Windows Defender event log analysis
- Windows Event ID review
- Endpoint detection validation
- SIEM event investigation
- PowerShell-based verification
- Containment and remediation documentation
- Professional incident reporting

Portfolio Summary

This scenario demonstrates the ability to validate endpoint protection telemetry, configure Wazuh to ingest a new event source, investigate a Defender detection in the SIEM, and verify remediation using PowerShell. This provides employer-facing proof of hands-on SOC triage, endpoint monitoring, and incident documentation skills.

Resume/Portfolio Bullet Draft

- Validated Microsoft Defender event ingestion in a Wazuh SIEM lab by generating a safe EICAR antivirus test detection, reviewing Defender Event ID 1117 locally, investigating the alert in Wazuh, and verifying quarantine/remediation through PowerShell.

Appendix A - Evidence Screenshots

The screenshots below document the detection workflow and are included as supporting evidence for the lab scenario.

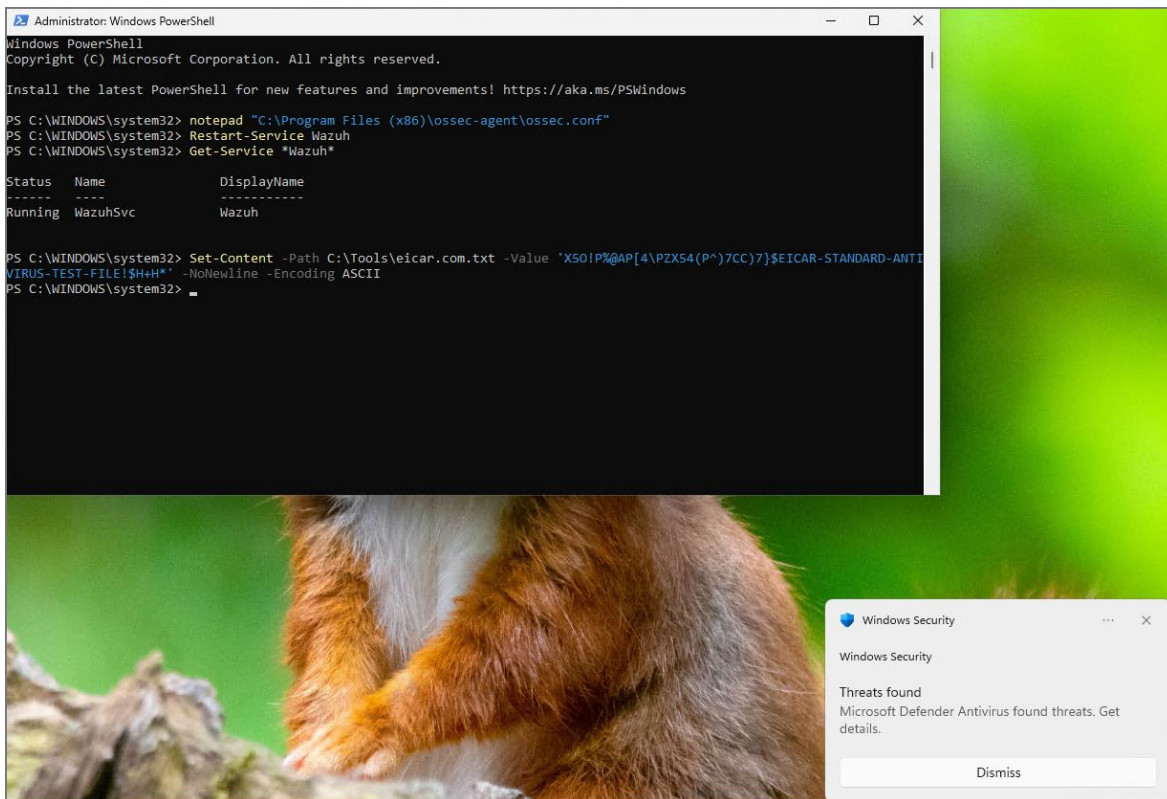


Figure 1. EICAR test file creation command and Microsoft Defender threat notification.

```
Administrator: Windows PowerShell
select {$_.Message -match "EICAR|Antivirus|threat|detected"} | Select-Object TimeCreated, Id, ProviderName, Message -First 5 | Format-List

TimeCreated : 5/18/2026 11:15:50 AM
Id           : 1117
ProviderName : Microsoft-Windows-Windows Defender
Message      : Microsoft Defender Antivirus has taken action to protect this machine from malware or other potentially
               unwanted software.
               For more information please see the following:
               https://go.microsoft.com/fwlink/?linkid=37020&name=Virus:DOS/EICAR_Test_File&threatid=2147519003&enterprise=0
               Name: Virus:DOS/EICAR_Test_File
               ID: 2147519003
               Severity: Severe
               Category: Virus
               Path: file:_C:\Tools\eicar.com.txt
               Detection Origin: Local machine
               Detection Type: Concrete
               Detection Source: Real-Time Protection
               User: NT AUTHORITY\SYSTEM
               Process Name: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
               Action: Quarantine
               Action Status: No additional actions required
               Error Code: 0x00000000
               Error description: The operation completed successfully.
               Security intelligence Version: AV: 1.449.681.0, AS: 1.449.681.0, NIS: 1.449.681.0
               Engine Version: AM: 1.1.26040.8, NIS: 1.1.26040.8
```

Figure 2. Local Microsoft Defender Event ID 1117 showing Virus:DOS/EICAR_Test_File, severe classification, and quarantine action.

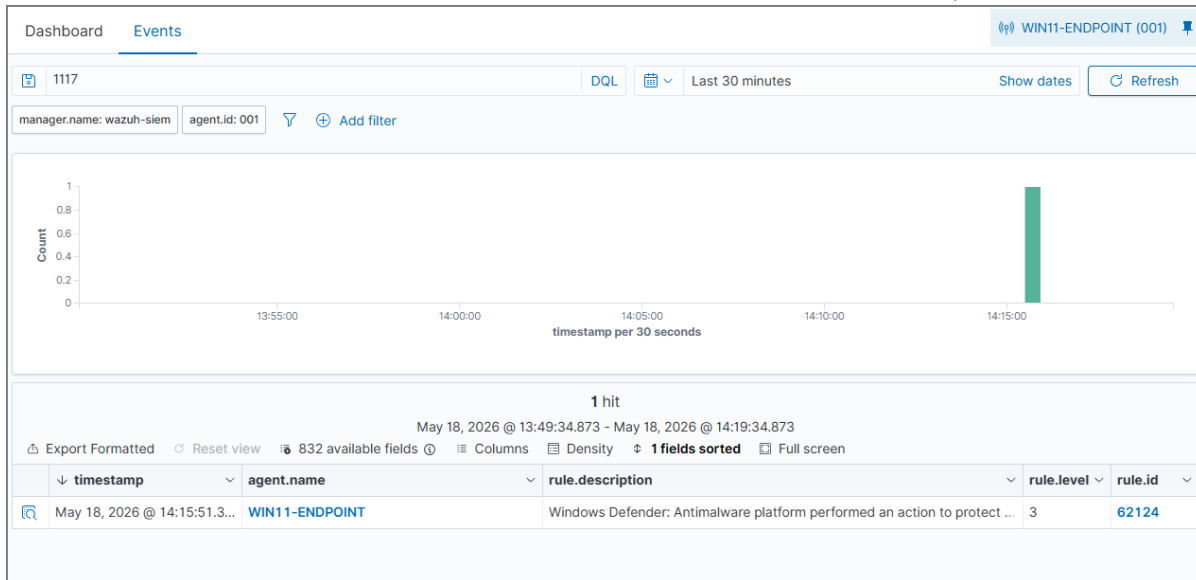


Figure 3. Wazuh Threat Hunting overview showing a Defender event from WIN11-ENDPOINT with rule ID 62124.

Document Details		view surrounding documents	view single document
data.win.eventdata.severity	Severe		
data.win.eventdata.source ID	3		
data.win.eventdata.source Name	Real-Time Protection		
data.win.eventdata.state	2		
data.win.eventdata.status Code	3		
data.win.eventdata.threat ID	2147519003		
data.win.eventdata.threat Name	Virus:DOS/EICAR_Test_File		
data.win.eventdata.type ID	0		
data.win.eventdata.type Name	Concrete		
data.win.system.channel	Microsoft-Windows-Windows Defender/Operational		
data.win.system.computer	WIN11-ENDPOINT		
data.win.system.eventID	1117		
data.win.system.eventRecordID	314		
data.win.system.keywords	0x8000000000000000		
data.win.system.level	4		
data.win.system.message	Microsoft Defender Antivirus has taken action to protect this machine from malware or other potentially unwanted software. For more information please see the following: https://go.microsoft.com/fwlink/?linkid=37020&name=Virus.DOS.EICAR_Test_File&threatid=2147519003		
data.win.system.opcode	0		
data.win.system.processID	3672		
data.win.system.providerGuid	{11cd958a-c507-4ef3-b3f2-5fd9dfbd2c78}		
data.win.system.providerName	Microsoft-Windows-Windows Defender		
data.win.system.severityValue	INFORMATION		
data.win.system.systemTime	2026-05-18T18:15:50.1323918Z		
data.win.system.task	0		
data.win.system.threadID	2708		
data.win.system.version	0		
decoder.name	windows_eventchannel		
id	1779128151.462445		
input.type	log		
location	EventChannel		
manager.name	wazuh-siem		
rule.description	Windows Defender: Antimalware platform performed an action to protect you from potentially unwanted software ()		

Figure 4. Wazuh document details showing Defender Operational channel, Event ID 1117, and threat name Virus:DOS/EICAR_Test_File.

```
PS C:\WINDOWS\system32> Test-Path C:\Tools\eicar.com.txt
False
PS C:\WINDOWS\system32> Get-MpThreatDetection | Select-Object -First 3 ThreatName,ActionSuccess,Resources,InitialDetectionTime

ThreatName ActionSuccess Resources InitialDetectionTime
-----
True {file:_C:\Tools\eicar.com.txt} 5/18/2026 11:15:39 AM

PS C:\WINDOWS\system32>
```

Figure 5. Cleanup verification showing the EICAR file no longer exists and Defender action success is True.